Cambridge Assessment International Education

Cambridge O Level

CHEMISTRY **5070/22**

Paper 2 Theory **May/June 2022**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2022 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

This document consists of **12** printed pages.

© UCLES 2022

[Turn over

PUBLISHED

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

© UCLES 2022

GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

© UCLES 2022

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	'List rule' guidance For questions that require ***n*** responses (e.g. State **two** reasons …): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked *ignore* in the mark scheme should not count towards ***n***. • Incorrect responses should not be awarded credit but will still count towards ***n***. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should **not** be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first ***n*** responses may be ignored even if they include incorrect science.

© UCLES 2022

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

© UCLES 2022

Question	Answer	Marks
1(a)	$KMnO_4$	1
1(b)	$Ba(NO_3)_2$	1
1(c)	KI	1
1(d)	AgCl	1
1(e)	Na_3N	1
1(f)	$KMnO_4$	1

Question	Answer	Marks
2(a)	2, 8, 6	1
2(b)	any value between 6.8 to 9.4 inclusive (g / cm^3)	1
2(c)	liquid because melting point is below 200 °C and boiling point is above 200 °C / 200 °C is between the melting point and the boiling point	1
2(d)(i)	correct dot-and-cross diagram	1
2(d)(ii)	weak intermolecular forces / intermolecular forces are easy to break or overcome	1
2(e)(i)	SeO_2 / O_2Se	1
2(e)(ii)	red / orange / yellow / yellow-green because non-metal oxides (are acidic)	1

© UCLES 2022

Question	Answer	Marks
2(f)	moles of oxygen = 0.94375 **OR** 30.2 / 32 (1) volume = 0.94375 × 24 **OR** 22.65 (1) volume = 23 (dm^3) (1)	**3**

Question	Answer	Marks
3(a)	**A** hydrogen (1) **B** magnesium butanoate (1) **C** carbon dioxide (1) **D** water (1)	**4**
3(b)	ethyl butanoate (1) $CH_3CH_2CH_2CO_2CH_2CH_3$ (1)	**2**
3(c)	does not completely dissociate / partially dissociates	**1**

Question	Answer	Marks
4(a)	79	**1**
4(b)	36	**1**
4(c)	$^{41}_{20}Ca$	**1**
4(d)	loses **two** electrons	**1**

© UCLES 2022

Question	Answer	Marks
4(e)(i)	in solid ions cannot move / in solid ions are in a lattice / in solid ions are fixed (1) in molten liquid ions can move (1)	**2**
4(e)(ii)	high melting point / high boiling point (1) dissolves in water (1)	**2**

Question	Answer	Marks
5(a)(i)	more ethanol (1) (equilibrium shifts to the right) to release energy / to release heat (1)	**2**
5(a)(ii)	less ethanol (1) (position of equilibrium shifts to the left) more of moles of **gas** on reactant side of equation / less moles of **gas** on product side (1)	**2**
5(b)	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ (1) **and any two from:** presence of yeast / enzyme (1) presence of water (1) quoted temperature between 15 and 45 (^oC) inclusive (1) absence of oxygen (1)	**3**

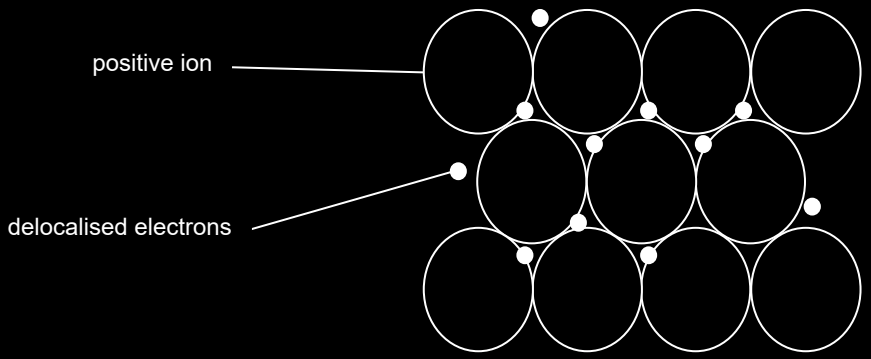
© UCLES 2022

Question	Answer	Marks
6(a)	$2Al(s) + 6H^+(aq) \rightarrow 2Al^{3+}(aq) + 3H_2(g)$ balancing (1) correct state symbol dependent on formulae (1)	**2**
6(b)	add (aqueous) sodium hydroxide (1) white ppt that redissolves / white ppt. soluble in excess (1) **OR** add (aqueous) ammonia (1) white precipitate (1)	**2**
6(c)	M_r for aluminium sulfate is 342 (1) x = 18 (1)	**2**
6(d)	**mixture** of a metal with another element	**1**

Question	Answer	Marks
7(a)	global warming / greenhouse effect / climate change	**1**
7(b)	acid rain / damage to buildings / killing living things	**1**
7(c)(i)	to increase the surface area (1) more exposed particles / more collisions per second (1)	**2**
7(c)(ii)	**particles** move slower / **particles** have less **kinetic** energy (1) fewer successful collisions / fewer collisions above that of activation energy (1)	**2**
7(d)(i)	(sulfur or sulfur dioxide) gains oxygen	**1**

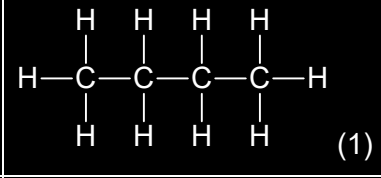
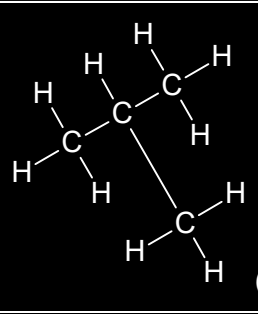
© UCLES 2022

Question	Answer	Marks
7(d)(ii)	bond making releases energy and bond breaking absorbs energy (1) more energy released than absorbed (1)	**2**
7(d)(iii)	lowers the activation energy	**1**

Question	Answer	Marks
8(a)	zinc is more reactive than iron / zinc is above iron in the reactivity series (1) zinc will react in preference to iron / zinc will oxidise instead of iron (1)	**2**
8(b)(i)	 diagram showing closely packed positive ions surrounded by delocalised electrons (1) **strong** attraction between electrons and positive ions (1)	**2**
8(b)(ii)	electrons can move / has mobile electrons	**1**
8(c)	cathode: $Zn^{2+} + 2e^- \rightarrow Zn$ (1) anode: $4OH^- - 4e^- \rightarrow O_2 + 2H_2O$ / $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$ (1)	**2**

© UCLES 2022

Question	Answer	Marks
8(d)	moles of zinc = 0.036 (1) moles of HCl = 0.100 (1) 0.1 mole of acid would react with 0.05 mol of zinc / 0.036 mole of zinc would need 0.072 moles of HCl (1)	**3**

Question	Answer	Marks
9(a)	 (1)  (1)	**2**
9(b)	$C_{12}H_{26}$	**1**
9(c)	**petroleum** is heated / **petroleum** is vapourised / **petroleum** is at high temperature (1) use of a (tall fractionating) column (1) separation depends on boiling point / high boiling point exits from the bottom / low boiling point from the top (1)	**3**
9(d)(i)	UV / sunlight	**1**
9(d)(ii)	amount of C is 37.8 / 12 (mol), amount of H is 6.30 / 1 (mol) and of Cl is 55.9 / 35.5 **OR** mole ratio C:H is 3.15 : 6.30 : 1.57 (1) empirical formula is C_2H_4Cl (1) molecular formula $C_4H_8Cl_2$ (1)	**3**

© UCLES 2022

Question	Answer	Marks
10(a)	**4** NH_3 + **5** O_2 → **4** NO + **6** H_2O	**1**
10(b)(i)	potassium hydroxide	**1**
10(b)(ii)	idea of titrating acid and alkali together using an indicator to find the end-point (1) repeat the procedure using the same volumes but without the indicator (1)	**2**
10(b)(iii)	M_r is 101 (1) % is 13.861 (1)	**2**
10(c)(i)	pesticides / herbicides / heavy metal **ions** / detergents	**1**
10(c)(ii)	chlorination (1) filtration / sedimentation / screening (1) use of carbon (1)	**3**

© UCLES 2022